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Project Synopsis

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TRIAGEAI

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ABSTRACT

The TriageAI project is an advanced, AI-powered Hospital OPD Queue Management System designed to revolutionize the traditional first-come-first-served patient queuing model used in outpatient departments across India and worldwide. The system leverages Artificial Intelligence (AI), Natural Language Processing (NLP), and real-time data analytics to automatically assess, score, and prioritize patients based on the medical urgency of their reported symptoms, ensuring that the most critical cases receive immediate attention regardless of their arrival time.

TriageAI introduces a comprehensive digital platform incorporating a patient-facing symptom submission interface, an AI-driven triage engine capable of generating severity scores and classifying patients as CRITICAL, HIGH, MEDIUM, or LOW priority, and a doctor dashboard that provides real-time clinical insights, risk scores, and red flag indicators for each queued patient. When a CRITICAL patient is detected, the system instantly alerts the concerned doctor, enabling swift medical intervention. The platform further supports a hospital administration panel that enables doctor registration, live queue monitoring, and hospital load tracking across departments.

Unlike conventional OPD management systems, which often result in long waiting times for seriously ill patients and inefficient utilization of medical resources, TriageAI eliminates these shortcomings by dynamically re-sorting the patient queue in real time as new patients are admitted. The system is built on a modern, scalable full-stack architecture using Java Spring Boot for the backend, Python FastAPI for the AI microservice, React or Angular for the frontend, and MySQL/MongoDB for persistent data storage. It is designed to comply with healthcare data privacy regulations and supports containerized deployment using Docker and Kubernetes for high availability.

Through its intelligent triage capabilities, TriageAI significantly reduces waiting times for critical patients, improves doctor workflow efficiency, reduces patient anxiety through transparent queue status visibility, and enables smarter hospital load management. The project aligns with the broader objective of modernizing healthcare infrastructure through technology

and sets a scalable, replicable precedent for AI-driven clinical decision support systems in hospital environments.

INTRODUCTION

The rapid advancement of Artificial Intelligence and digital technologies has created unprecedented opportunities to transform healthcare delivery systems globally. Hospital Outpatient Departments (OPDs) are among the most critical touchpoints in a healthcare system, serving as the first point of contact for millions of patients seeking medical attention. However, traditional OPD management systems continue to operate on a first-come-first-served queuing model that fails to account for the varying degrees of medical urgency among patients. This fundamental flaw results in critical patients waiting alongside non-urgent cases, often leading to delayed care, adverse health outcomes, and increased patient mortality in severe cases.

In India, the burden on hospital OPDs is particularly severe, with a large patient population relative to the number of available physicians and medical facilities. The absence of systematic triage at the OPD level means that clinical staff must manually assess and prioritize patients, a process that is time-consuming, inconsistent, and prone to human error. Furthermore, patients have little or no visibility into their queue status or expected wait times, contributing to anxiety, dissatisfaction, and congestion in hospital waiting areas.

In response to these challenges, the TriageAI – AI-Based Hospital OPD Queue Management System proposes a holistic, technology-driven solution that integrates AI-powered symptom analysis with a real-time dynamic queue management framework. The system enables patients to submit their symptoms via a text or voice interface, after which an intelligent AI triage engine processes the input using NLP and medical reasoning models to generate a structured clinical analysis. This analysis includes a numerical risk score, a severity classification (CRITICAL, HIGH, MEDIUM, or LOW), possible diagnoses, and red flag indicators that alert medical personnel to potentially life-threatening conditions.

The TriageAI platform consists of three primary user-facing modules: a patient interface for symptom submission and queue status monitoring, a doctor dashboard displaying AI-sorted patient queues with detailed clinical insights, and an administrative control panel for hospital operations management. Each of these modules communicates through a robust RESTful API architecture that ensures seamless data flow and real-time updates across the system.

TriageAI stands apart from existing hospital queue management tools by offering explainable AI triage reasoning, instant critical alerts to physicians, and a fully transparent patient experience. The platform is designed to be scalable, maintainable, and compliant with healthcare data security standards. Ultimately, TriageAI aspires to reduce preventable patient deterioration, optimize medical resource allocation, and create a safer, more efficient OPD environment for both patients and healthcare providers.

LITERATURE SURVEY

S.No	Author(s) & Year	Title of Paper / Work	Objective / Focus	Methodology / Approach	Key Findings
1	Rajpurkar et al., 2022	AI in Clinical Decision Support: A Systematic Review	Examine the effectiveness of AI-based clinical decision support tools in hospital environments	Systematic literature review of AI implementations across emergency and OPD settings globally	AI models significantly outperform traditional triage in accuracy, speed, and severity classification.
2	Chen et al., 2023	NLP-Driven Symptom Analysis for Patient Prioritization	Develop an NLP pipeline to analyze patient-reported symptoms and assign clinical priority	NLP model trained on anonymized patient records for symptom classification and triage scoring	NLP-based symptom analysis achieves over 88% accuracy in identifying high-severity cases.
3	Moreschini et al., 2024	Edge to Cloud Tools: A Multivocal Literature Review	Survey modern edge-cloud tools and their suitability for real-time healthcare applications	Synthesis of academic and industrial solutions in distributed application architectures	Edge-cloud integrations are highly effective for real-time, scalable healthcare platforms.
4	Anonymous, 2023 (IJRPR)	AI-Driven Hospital Queue Management System	Design and evaluate an AI-powered system for OPD queue management and critical patient detection	Web-based system integrating AI severity scoring and dynamic real-time queue re-sorting	AI-driven queue systems reduce critical patient wait times by up to 60%.
5	Mondal, S., 2024	Cloud Computing in Healthcare: Economic and Operational Impact	Explore cloud computing's effect on hospital operational efficiency and cost reduction	Narrative analysis of hospital cloud platform adoption focusing on scalability and data management	Cloud-based hospital apps reduce operational costs and support real-time clinical analytics.

PROBLEM STATEMENT

Despite significant advancements in medical science and hospital infrastructure, the management of patient queues in Outpatient Departments (OPDs) across Indian hospitals continues to rely predominantly on manual, first-come-first-served processes. This approach is fundamentally inadequate for handling diverse patient populations with varying levels of medical urgency. Critical patients with potentially life-threatening conditions are routinely made to wait alongside patients with minor ailments, resulting in delayed diagnosis, deteriorating health outcomes, and in extreme cases, preventable mortality.

Existing OPD queue management systems, where they exist, lack the intelligence required to distinguish between patients based on symptom severity. Medical staff are overburdened with manual triage responsibilities, leaving little time for focused clinical care. Furthermore, there is no standardized mechanism to alert doctors instantly when a high-risk patient enters the queue. Patients have no visibility into their queue position or expected waiting time, leading to anxiety, overcrowding in waiting areas, and frequent complaints about hospital service quality.

The core problem is the absence of a unified, AI-driven platform that can automatically assess patient symptoms, generate medically sound severity classifications, dynamically manage patient queues in real time, and provide clinical decision support to treating physicians. There is an urgent need for a system that bridges the gap between patient arrival and clinical prioritization through intelligent automation.

Objectives:

- Develop an AI-powered triage engine that analyzes patient-reported symptoms to generate severity scores and classify patients as CRITICAL, HIGH, MEDIUM, or LOW priority.
- Implement a real-time dynamic OPD queue management system that continuously re-sorts the patient queue based on medical urgency.
- Design a doctor dashboard that provides live AI-sorted patient queues with clinical summaries, risk scores, red flag indicators, and suggested diagnoses.
- Build a patient-facing interface enabling symptom submission via text or voice with real-time queue position and estimated wait time updates.
- Create an administrative control panel enabling hospital administrators to register doctors, monitor queue loads, and manage OPD configurations.

- Ensure instant critical patient alerts are delivered to the treating physician's dashboard when a CRITICAL severity patient is detected.
- Maintain comprehensive triage logs for auditing, monitoring, regulatory compliance, and AI model improvement.
- Design the platform for scalability, high availability, and compliance with healthcare data privacy standards using containerized deployment.

PROPOSED METHODOLOGY

The TriageAI platform adopts a modern, full-stack web and microservices development approach to deliver a scalable, intelligent, and user-centric OPD queue management system. The methodology encompasses a carefully selected technology stack, a modular system architecture, and well-defined data flow workflows that ensure secure, efficient, and real-time AI-driven patient triage and queue management.

Tech Stack

1. Frontend: React.js / Angular, TypeScript, Tailwind CSS – provides responsive, intuitive, and accessible user interfaces for patients, doctors, and administrators across all devices including kiosks, tablets, and mobile phones.
2. Backend: Java Spring Boot (v3+) – manages RESTful API endpoints, business logic, authentication, and real-time queue management operations.
3. AI Microservice: Python FastAPI – hosts the core AI triage engine responsible for NLP-based symptom analysis, risk score generation, severity classification, red flag detection, and clinical reasoning generation.
4. Database: MySQL (patient triage records, queue data) and MongoDB (session logs, audit trails) – provides robust relational and document-based storage with support for real-time data retrieval.
5. AI Engine: OpenAI API / Local NLP Model (GPT Medical NLP) – drives intelligent symptom interpretation, clinical reasoning, and explainable AI-based triage decisions.
6. Infrastructure: Docker, Kubernetes, API Gateway, Load Balancer, CDN – supports containerized deployment for scalability, fault tolerance, and high availability.

TriageAI – System Flowchart

Patient Triage & Queue Management Workflow

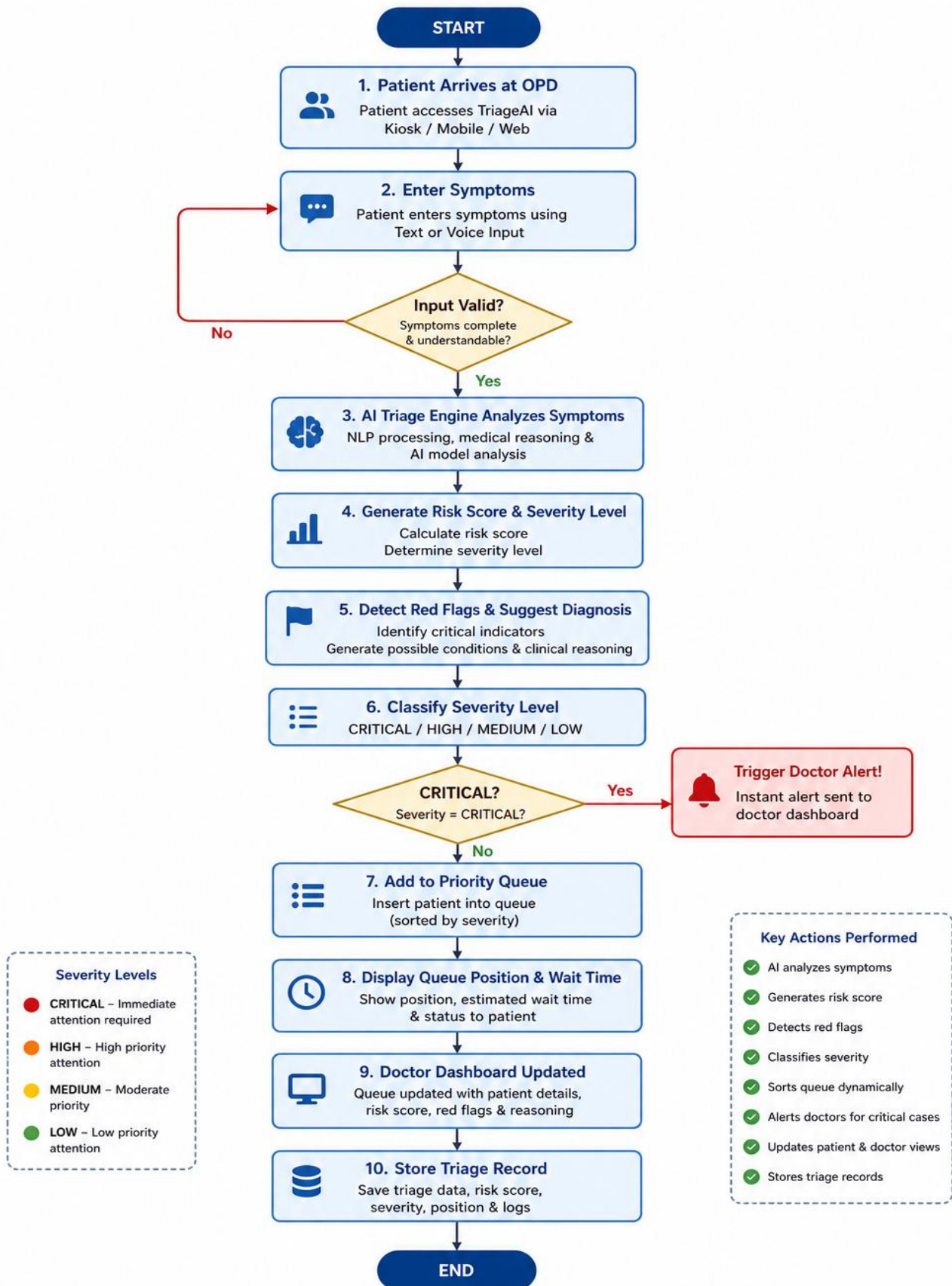


Fig 1: TriageAI System flowchart

Block Diagram

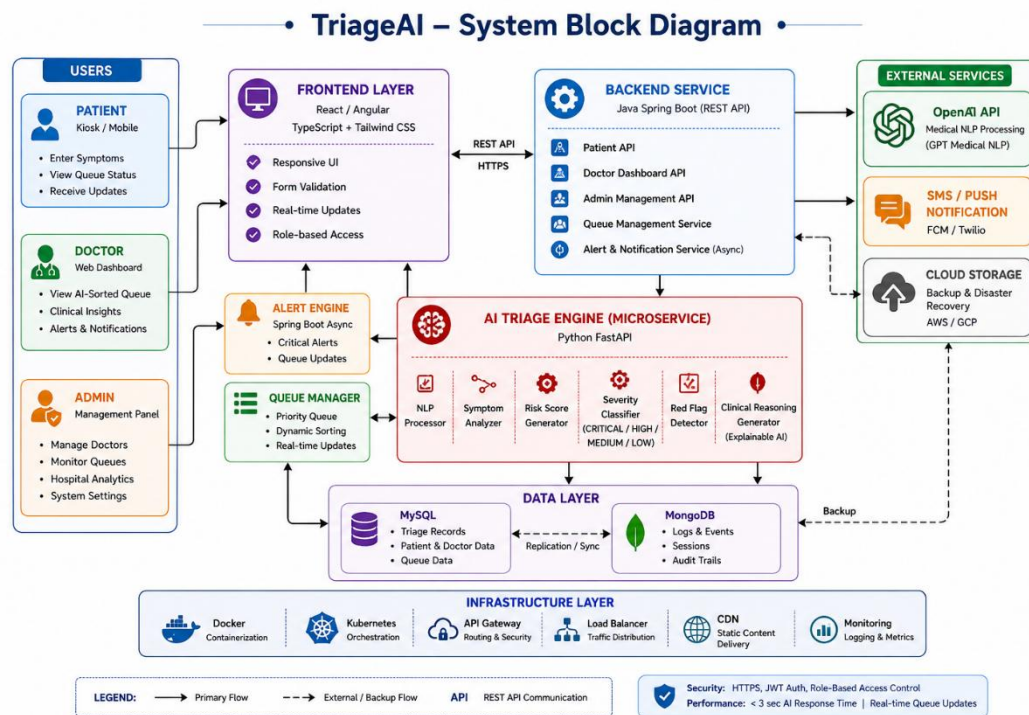


Fig 2: TriageAI System Block Diagram

Key Features Enabling the Methodology

- AI-powered symptom analysis with explainable clinical reasoning and red flag detection.
- Real-time dynamic queue management with continuous priority re-sorting.
- Instant critical patient alerts delivered directly to the doctor's dashboard.
- Transparent patient queue status with live position and estimated wait time.
- Secure data management with role-based access control and HIPAA-compliant data handling.
- Containerized deployment using Docker and Kubernetes for scalability and high availability.

This methodology ensures that TriageAI delivers a reliable, intelligent, and scalable solution for modern hospital OPD management, fostering patient safety and maximum clinical efficiency at every stage of the triage process.

SYSTEM DIAGRAMS

Use Case Diagram

The Use Case Diagram below illustrates the interactions between the three primary actors – Patient, Doctor, and Hospital Admin – and the core functionalities of the TriageAI OPD Queue Management System. The system boundary encloses all use cases handled internally by TriageAI, including AI triage engine analysis, queue management, and critical alert generation.

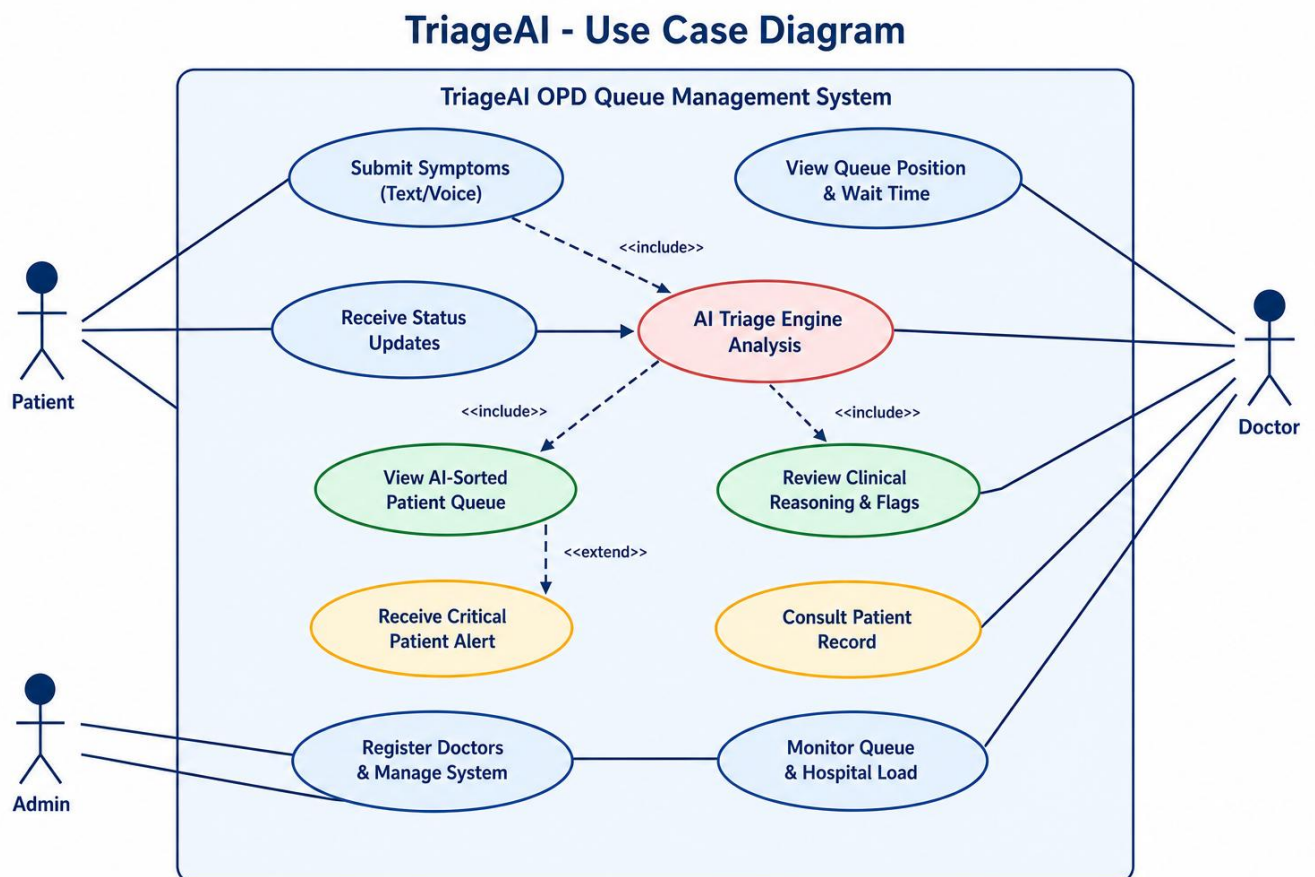


Fig 3: TriageAI Use Case Diagram

Sequence Diagram

The Sequence Diagram below describes the time-ordered interaction between system components during the primary scenario of patient triage and queue assignment. It captures how messages flow from the patient through the TriageAI system, the AI Triage Engine, the Queue Manager, the Doctor, and the Database to complete a full triage cycle, including critical patient alerting.

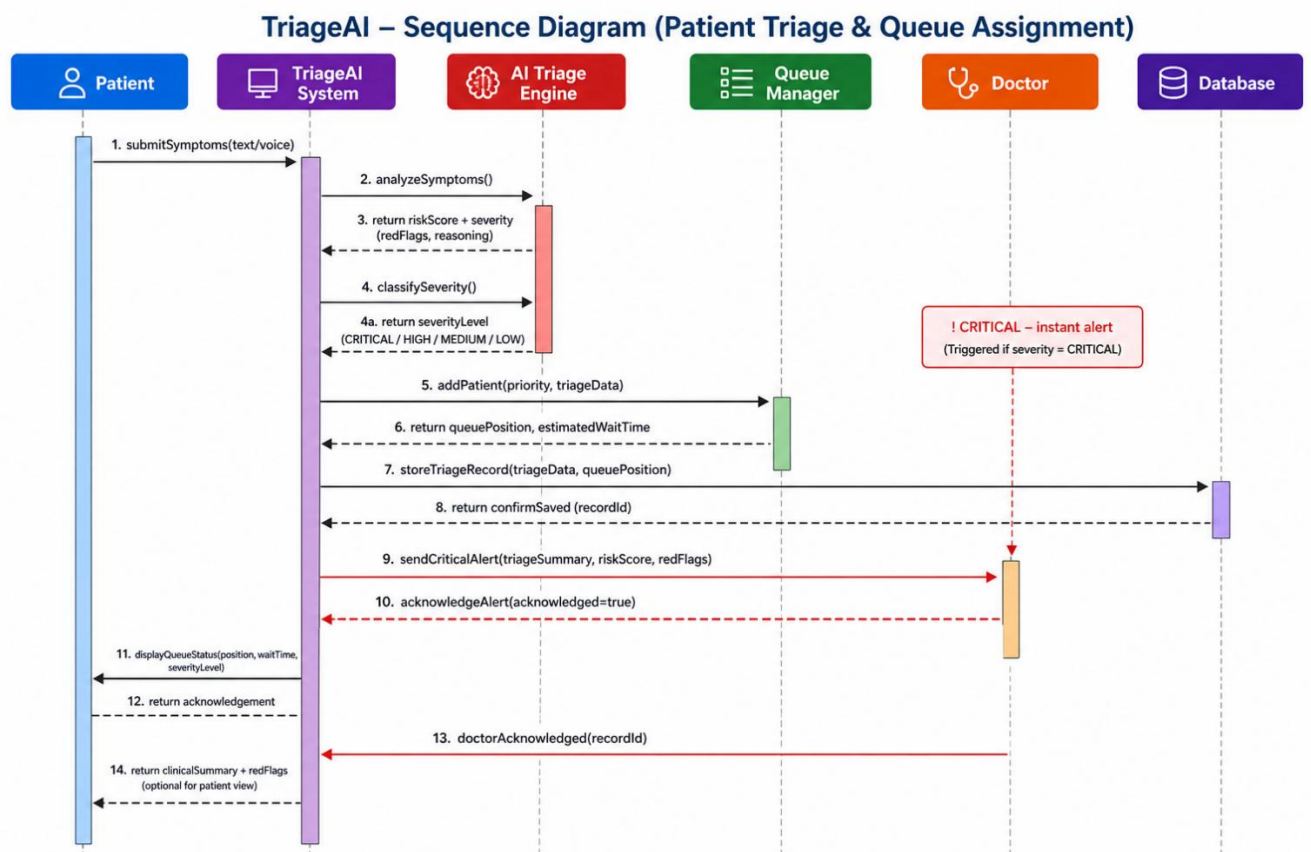


Fig 4: TriageAI Sequence Diagram – Patient Triage & Critical Alert

CONCLUSION AND FUTURE SCOPE

Conclusion

The TriageAI – AI-Based Hospital OPD Queue Management System presents a transformative approach to one of the most persistent challenges in modern healthcare: the inefficient and potentially dangerous practice of managing OPD patient queues without clinical prioritization. By leveraging Artificial Intelligence, Natural Language Processing, and real-time queue management technologies, TriageAI ensures that patients with the most critical medical needs receive attention first, dramatically reducing the risk of adverse health outcomes due to delayed care.

Through its integrated platform – encompassing an AI-driven triage engine, a real-time dynamic queue management system, a clinician-facing dashboard with intelligent insights, a transparent patient interface, and a comprehensive administrative control panel – TriageAI addresses the systemic failures of conventional OPD management. The architecture, built on proven enterprise technologies including Java Spring Boot, Python FastAPI, MySQL, MongoDB, Docker, and Kubernetes, ensures that the platform is scalable, maintainable, and ready for production deployment in real hospital environments.

TriageAI's ability to generate explainable AI triage decisions, detect critical red flag symptoms, and deliver instant physician alerts represents a significant advancement over existing queue management solutions. The system's comprehensive triage logging further supports regulatory compliance, audit readiness, and continuous AI model improvement. In summary, TriageAI provides a viable, technology-enabled standard for intelligent OPD management that prioritizes patient safety above all else.

Future Scope

Looking ahead, the TriageAI platform has substantial potential for enhancement, expansion, and deeper integration within the broader healthcare ecosystem. The following future developments are envisioned:

- **Mobile Application Development:** A dedicated Android and iOS mobile application will be developed to allow patients to submit symptoms remotely before arriving at the hospital, further reducing in-facility congestion and wait times.

- **Advanced AI Model Integration:** Future iterations of the triage engine will incorporate fine-tuned large language models specifically trained on clinical datasets to improve diagnostic accuracy and the quality of explainable clinical reasoning.
- **Integration with Hospital EMR/EHR Systems:** The platform will be extended to interface with Electronic Medical Record systems via HL7/FHIR APIs, enabling the AI engine to leverage complete patient medical histories for more accurate triage.
- **Wearable Device Integration:** Future versions will support real-time health data ingestion from patient wearables such as smartwatches and fitness trackers, enabling physiological parameters to be factored into the AI triage assessment.
- **Blockchain for Immutable Triage Records:** The addition of a blockchain-based audit trail will ensure that all triage records are immutable, tamper-proof, and fully transparent, strengthening medico-legal compliance.
- **Multilingual Interface Support:** To serve India's diverse linguistic population, the platform will be enhanced with multilingual support, allowing patients to submit symptoms in their preferred regional language.
- **Telemedicine Integration:** Future development may incorporate a telemedicine module, allowing doctors to conduct preliminary virtual consultations with high-priority patients before their physical appointment.
- **Predictive Analytics and Hospital Load Forecasting:** Advanced predictive analytics will be integrated to forecast daily OPD patient volumes, identify peak hours, and assist administrators in proactive resource allocation and staffing planning.

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